

Mihai Sorin Dobre

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INTERESTS

- *Generative AI*: Agentic systems, Retrieval-Augmented Generation, LLM applications
- *Machine Learning*: Deep Learning, (Multi-agent) Reinforcement Learning, Imitation Learning, Bayesian Methods
- Multi-agent Systems and Game Theory
- *Applications*: Regulatory compliance for MedTech; Perception, Prediction and Planning for Autonomous Driving

PROFESSIONAL EXPERIENCE

Co-founder & CTO
Klaris AI, UK

Dec 2024 – present
klaris.ai

- Co-founded Klaris AI to automate regulatory compliance for medical devices with AI agents that review technical documentation against expert-validated requirements
- Drove applied research across MedTech regulation and frontier AI: evaluated leading models and methods to continually push performance on real regulatory documents
- Built an evaluation framework on public and internal data to benchmark state-of-the-art approaches and systematically tune our systems
- Developed the in-house AI system, including custom retrieval, knowledge graphs, agentic workflows and integrated OCR engines, to maximise performance on the heterogeneous, complex documents typical of regulatory submissions
- Single-handedly owned the backend, database and infrastructure while coordinating out-sourced frontend engineers to ship the platform; the product has served dozens of customers and pilots
- Achieved ISO/IEC 27001:2022 certification and successfully passed the audits; defined security, development, CI/CD and documentation processes
- Hired and now lead a small engineering team; own the product roadmap across technical delivery and customer-facing priorities
- In close collaboration with my co-founder, raised \$1M to launch Klaris from idea to a product that can scale to serve customers across multiple jurisdictions

Entrepreneur in Residence
Antler, London, UK

Sep 2024 – Dec 2024

- Completed the Antler incubator programme and secured funding to found Klaris AI
- Partnered with co-founder Francesco Corazza to validate the MedTech regulatory compliance opportunity

Lead Research Scientist
Five AI / Bosch, UK

May 2023 – Jul 2024

- Led the Motion Planning and Prediction applied research team: technical direction, cross-org collaboration with Bosch and academia, and line management
- Research on multi-agent interactions, robust planning under distribution shift, and tighter integration of prediction with planning; hybrid ML (GNNs, Transformers, MARL, imitation) with classical methods (Iterative Best Response, Bayesian inference)

- Delivered ML solutions for HD Map Reconstruction from camera and LiDAR for autolabelling; guided engineering to bring research into the deployed vehicle stack
- Extended the research simulator to multi-agent settings and accelerated it with JAX (vectorisation, JIT, hardware acceleration)

Senior Research Scientist in Motion Planning and Prediction
Five AI, UK

May 2020 – May 2023

- Led Interactive Prediction and Planning: MCTS ego planning with sampled predictions, and multi-agent models of simultaneous interaction
- Built prediction via neural models with Bayesian goal recognition; occlusion handling via optimisation-based planning and belief modelling (e.g. particle filtering)
- Developed evaluation frameworks and metrics for prediction and planning to guide product priorities

Research Scientist in Motion Planning and Prediction
Five AI, UK

May 2018 – May 2020

- Designed a high-level planner for complex manoeuvres (e.g. roundabout merging, overtaking) and a research driving simulator used across RL, interactive planning and prediction work

Software Developer on STAC project
University of Edinburgh, UK

Sep 2014 – Mar 2015 & Aug – Dec 2016

- Collaborated with Toulouse and Heriot-Watt on a rule-based Settlers of Catan agent and ML methods for negotiation

Software Developer Intern on WebSphere Application Server
IBM, Hursley, UK

Jul 2011 – Aug 2012

SELECTED PUBLICATIONS

- Aron D, Yiwei W, Faris J, Marcel H, Mihai D, Alexander L, Joschka B, and Johannes B. Driving is a game: Combining planning and prediction with bayesian iterative best response. In *ECCV*, Malmö, Sweden, 2026. to appear
- Morris A*, Mihai D*, Stefano A, John R, and Subramanian R. Flash: Fast and light motion prediction for autonomous driving with bayesian inverse planning and learned motion profiles. In *IEEE IROS*, Kyoto, Japan, 2022
- Josiah H, Arrasy R, Elliot F, Francisco E, Mihai D, John R, Subramanian R, and Stefano A. Interpretable goal recognition in the presence of occluded factors for autonomous vehicles. In *IEEE IROS*, Prague, Czech Republic, 2021
- Stefano A, Cillian B, John W, Balint G, Francisco E, Mihai D, and Subramanian R. Interpretable goal-based prediction and planning for autonomous driving. In *IEEE ICRA*, Xi'an, China, 2021
- Mihai Dobre and Alex Lascarides. POMCP with human preferences in settlers of catan. In *Proceedings of the Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, Edmonton, Canada, 2018
- Joachim F, Ben K, Mihai D, Marco D, Emmanuel K, Daniel D, Bonnie W, and Federico F. Talking to myself: self-dialogues as data for conversational agents. *CoRR*, abs/1809.06641, 2018
- Ben K, Marco D, Mihai D, Daniel D, Joachim F, Federico F, Emmanuel K, Jianpeng C, and Bonnie L. W. Edina: Building an open domain socialbot with self-dialogues. *CoRR*, abs/1709.09816, 2017

EDUCATION

PhD, Informatics

2013 – 2019

Low-resource learning in complex games

University of Edinburgh, UK

Supervisors: Alex Lascarides, Subramanian Ramamoorthy

BSc (Hons) Robotics with Artificial Intelligence (1st class)

2009 – 2013

University of Bradford, UK

Final year project: Modelling a Hide and Seek Game under Uncertainty

SELECTED COMPUTING SKILLS

Languages: Python (NumPy, pandas, etc.), C++, Java, SQL

AI / ML: LLMs, agentic workflows, custom retrieval / RAG, Torch, TensorFlow, JAX, RLLib

Infrastructure: AWS, Azure, Docker, CI/CD, Git

Other: ROS, Unix Shell, L^AT_EX

AWARDS

- Best Student Paper Award for “Exploiting action categories in learning complex games”
- Best Overall Performance Award received for BSc degree (2013)

PREVIOUS PROJECTS AND EXPERIENCE

- *Amazon Alexa Challenge 2017:* socialbot for open-domain conversation with team Edina (University of Edinburgh).
- *Strategic Conversation (STAC):* models combining linguistic theory, agent interaction and decision making.
- Teaching assistant at Edinburgh (Reasoning and Agents; Reinforcement Learning) and Bradford (Software Development), including lecturing, labs and assessment for large cohorts.

OTHER INTERESTS

Tennis: competed on the ITF World Pro Circuits, on Romanian and British Tour, and University team levels.